Md Modassir Firdaus

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Education

PhD Candidate in Mechanical Engineering Jan-2020 to Ongoing

Indian Institute of Technology Gandhinagar (IITGN),
Visiting PhD Scholar, BART Lab Mahidol University

Thesis: "*Virtual Actuation Framework for Kinematics and Control of Tendon-Driven Continuum Robots with Cadaveric Validation in Colon Navigation*" Thesis Submitted

Thesis Adviser: Prof. Madhu Vadali (Mechanical Engineering IITGN, India)

Surgical Robot Development Adviser: Prof. Jackrit Suthakorn (BART Lab Mahidol University, Thailand) and Prof. Branesh M. Pillai (AIT, Thailand)

Clinical Trial Adviser: Prof. Chumpon Wilasrusmee (Minimally Invasive Surgeon, Ramathibodi Hospital, Thailand)

Bachelors in mechanical engineering July 2012- April 2016

Jawaharlal Nehru Technological University
75.4 %, Best CGPA of the batch

Founder and Faculty of Physics and Mathematics May 2016 -December 2019

Firdaus Academy for JEE Main & Adv.

- I have learned to manage an institute.
- Leading a team of faculties to fulfill the objective.

Learn to deal with and manage the challenges created by students to a different extent.

Publications

[J1]	**Md Modassir Firdaus**, Omkar Dilip Paranjape, and Madhu Vadali	Finite Element Method-Based Dynamic Modeling Framework for Flexible Continuum Manipulators	ASME Journal of Mechanisms Robotics [Q1]
[C2]	**Md Modassir Firdaus** and Madhu Vadali	Virtual Tendon-Based Inverse Kinematics of Tendon-Driven Flexible Continuum Manipulators	AIR '23: Proceedings of the 2023 6th International Conference on Advances in Robotics, ACM [IF 1.3]
[C2]	**Md Modassir Firdaus**, Smit Parjapati, and Madhu Vadali	Initial State Update Based Position Control of Flexible Continuum Manipulators Utilizing Virtual Actuation Space	IEEE International Conference on Robotics and Biomimetics (IEEE-ROBIO 2024) [IF 1.81]
[J2]	**Md Modassir Firdaus**, Shail Jadav, and Madhu Vadali	Trajectory Tracking of Multi-Section Tendon-Driven Flexible Continuum Manipulators using Virtual Actuation Space Control.	Robotica (Accepted)
[J3]	**Md Modassir Firdaus**, B. M. Pillai, Madhu Vadali, C. Wilasrusmee and J. Suthakorn	Design and Cadaveric Evaluation of a Low-Cost, 3D-Printed Tendon-Driven Continuum Endoscope for Colon Intervention: A Preliminary Study	NJP| Robotics (Under Review)

[C3]	**Md Modassir Firdaus**, Vikranth Mallru, Harsh Malodia, and Madhu Vadali	Bioinspired Rolling-Disk Continuum Robot: Logarithmic Spiral and Constant Curvature Design with Contraction Capabilities	ICRA 2026 (Submitted)
[J4]	**Md Modassir Firdaus** and Madhu Vadali	Topology Informed Point Claud Based Path Planning Using RRT* for Anatomical Cavity and Different Topological Space.	IJRR (Submitted)
[J5]	**Md Modassir Firdaus** and Madhu Vadali	Autonomous Confined Cavity Navigation with Combined Kinematics of Tendon-Driven Continuum Robots and a Rigid Linked Manipulator Utilizing Virtual Actuation Space	(Under Preparation)

Patent

1. Vadali, Madhu and **Firdaus, Md Modassir**, "A system for performing minimally invasive surgery (MIS)", Indian Patent Office, Patent Application No.: 202421002550, (Published)
2. **Md Modassir Firdaus**, Harsh Malodia, and Madhu Vadali, "A Rolling Disk Embodied Design of Continuum Robot" || P-600780 (Filing)
3. **Md Modassir Firdaus,** and Madhu Vadali, "Ergonomic Flexible Minimally Invasive (ErgoFlex MIS) Instrument with Wrist-Aligned Gimbal Actuation" || P-600779 (Filing)

Experience as Reviewer

- IEEE APSCON
- IEEE RAL
- International Conference on Advances in Robotics
- i-Create 2025

Awards

[A1]	2025	**Winner of MIT Media Lab AI and Robotics Hackathon 2025**
[A2]	2025	**Gandhian Young Technological Innovation (GYTI) Award 2024, for the development of "Flexible Robot for Transanal Minimally Invasive Intervention"** (One of the most prestigious awards given by Office of the President of India at Rashtrapati Bhavan)
[A3]	2024	Overseas Research Exposure Fellowship by Indian Institute of Technology Gandhinagar
[A4]	2024	Mahidol University Residence Fellowship
[A5]	2024	Grant for Attending International Conference
[A6]	2024	Winner Regional Finale of Boeing BUILD
[A7]	2014	Winner college level line following mobile robot competition.

Talk and Seminar Delivered

IEEE Medical Robotics Seminar, Mahidol University, Thailand	Nov 06, 2024
Invited Speaker AI- Robotics Conference Dialogue on Healthcare MBZUAI Abu Dhabi	April 23, 2025- April 25, 2025

Skills

Control Dynamixel and Rigid Linked Robots like **xArm6**, **Adverb Heal**
Designing the continuum robot using **CAD**

Sensors, Actuators and Controller Integration (Electronic Setting)
Worked with **NI Based Real-Time Controller** and **DAQ**

Latex, Graphic Design

Motion Capture, Continuum Robot and Adverb Heal Integration using **ROS 1**
MATLAB for Mathematical Calculation and Simulations
Python and **Lab View** for Hardware Implementations
CNC, Lathe, Leaser Cutting, 3D Printing and other Manufacturing Tools for building experimental setup and prototypes

Teaching Assistant Experience

Graduate Teaching Fellow [Manufacturing]
Indian Institute of Technology Gandhinagar
Taught undergraduate course on Manufacturing Technology I
Prof. Madhu Vadali

Teaching Assistant
Control Theory
Indian Institute of Technology Gandhinagar
I taught undergraduate courses on control theory along with
Prof. Madhu Vadali

Teaching Assistant
ME LAB II: Enabled students to do "learning-by-doing"
Indian Institute of Technology Gandhinagar
Writing: Enabled students for scientific writing
Prof. Harish PM

Teaching Assistant
Manufacturing Technology I & II
Indian Institute of Technology Gandhinagar
Prof. Madhu Vadali

Teaching Assistant
Introduction to Control Theory
Indian Institute of Technology Gandhinagar
Prof. S. Rajendaran

Relevant Course Work

Introduction to Robotics
Control Theory
Modern Control Theory
Nonlinear Control
Mechanics of Rod
Digital Control Systems

Hobbies

Playing Squash| Swimming, Represented IIT Gandhinagar in inter-IIT swimming championship, Free-Style |

Some Extra Activity

Karate Yellow Belt | Annual Training Camp NCC, Army Wings | Reading the Holy Quraan